

Madhur N Patel

Infrastructure & Embedded Systems Engineer

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SUMMARY

Builds and operates self-managed Linux infrastructure, containerized services, and sensor-fusion systems on Raspberry Pi and ESP32. Looking for infrastructure, embedded, or platform engineering work — ideally somewhere that ships physical products.

PROJECTS

GPS-Guided Dual-Device Tracking System

github.com/pmadhurn/gps-guided-tracking-system

- Two Raspberry Pi-controlled gimbals point toward each other by GPS bearing, then hand off to computer vision for fine LED tracking; automatic mode switching between coarse GPS and camera tracking.
- Real-time 9-axis IMU650 sensor fusion (GPS, IMU, magnetometer), Haversine distance and elevation-angle calculation, SP2050 gimbal control under PID regulation.
- Multi-threaded serial and socket communication between the two devices over WiFi.
- Python, OpenCV, PySerial, Raspberry Pi 4/5.

Self-Hosted Infrastructure — OCI ARM64 + Cloudflare Tunnel

github.com/pmadhurn/OCI

- Self-managed production environment running six containerized services on a 4-core / 24GB ARM64 Oracle Cloud instance (Open WebUI, n8n, Portainer, code-server, Ollama, Supabase).
- Zero-trust ingress through Cloudflare Tunnel with no public inbound ports; subdomain routing via tunnel ingress rules with an ordered catch-all fallback.
- Local LLM inference on ARM64 via Ollama; self-hosted Supabase with credential isolation; service monitoring, backup, and TLS certificate management.
- Docker, Docker Compose, Nginx, Linux (Ubuntu), PostgreSQL.

SpeakInsights — Meeting Intelligence Platform

github.com/pmadhurn/SpeakinsightsV6

- Docker-deployable meeting platform: LiveKit WebRTC conferencing for up to 20 participants, WhisperX transcription with speaker attribution, Ollama summarization and task extraction, RAG chat over meeting context with pgvector.
- Runs fully on-premises — no third-party API calls, so meeting audio never leaves the deployment.
- Async FastAPI backend with a Redis-backed job queue; batched GPU inference for long-form audio; rolling container restarts via Docker Compose.
- React 18, TypeScript, FastAPI, Python 3.11, PostgreSQL + pgvector, Redis, Docker.

EXPERIENCE

Self-Directed Infrastructure Engineering

2022 – Present

- Implemented robust CI/CD pipelines using GitHub Actions, automating the build, test, and deployment of applications.
- Engineered a secure and scalable network infrastructure using Cloudflare Tunnel, enabling secure access to internal applications and services without exposing them to the public internet.
- Managed Vercel deployments with a non-proxied Cloudflare setup, optimizing for performance and security by leveraging Cloudflare's DNS and security features while directly serving content from Vercel.
- Containerized services with Docker and Docker Compose, managing service dependencies, health checks, and persistent volumes across a six-service deployment.

EDUCATION & CERTIFICATIONS

B.Tech, Computer Science and Engineering — Indus University, Ahmedabad

2022 – 2026

GPA 8.5/10

AWS Academy Graduate — Machine Learning Foundations (Aug 2025) • AWS Academy Graduate — Cloud Foundations (Jul 2025) • MERN Stack Developer, Coding Cloud (Apr 2025)

SKILLS

Strongest: Python, JavaScript, Node.js, React.js, Git/GitHub

Infrastructure: Docker, Docker Compose, Cloudflare Tunnel, Nginx, Oracle Cloud, Vercel, CI/CD, AWS, GCP

Embedded/IoT: Raspberry Pi, ESP32, Arduino, Serial / I2C • **Systems:** Linux (Ubuntu), macOS, Windows

Languages: Python, JavaScript, TypeScript, SQL, C, C++, Java, Bash

Backend: FastAPI, Node.js, Express.js, REST APIs • **Databases:** PostgreSQL, MongoDB, MySQL, SQLite, Redis

AI/ML: OpenCV, Ollama (local inference), Pandas, NumPy, TensorFlow, scikit-learn